

Departure From Predictability

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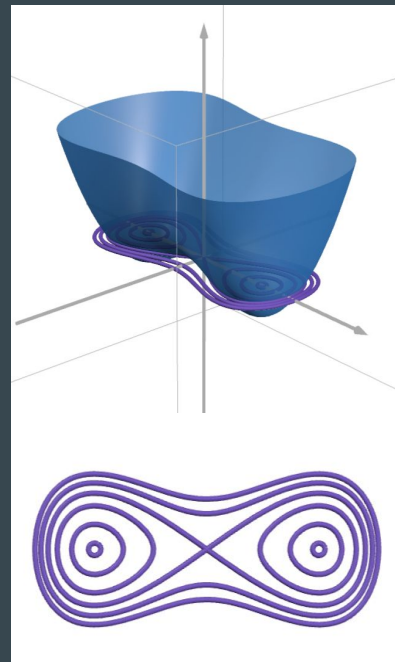
Chaos in a Modified Duffing Oscillator

Intro

- The Duffing Oscillator is a damped driven oscillator with a potential curve like the one shown on the right.
- The differential equation for how a point particle behaves in this system is given as

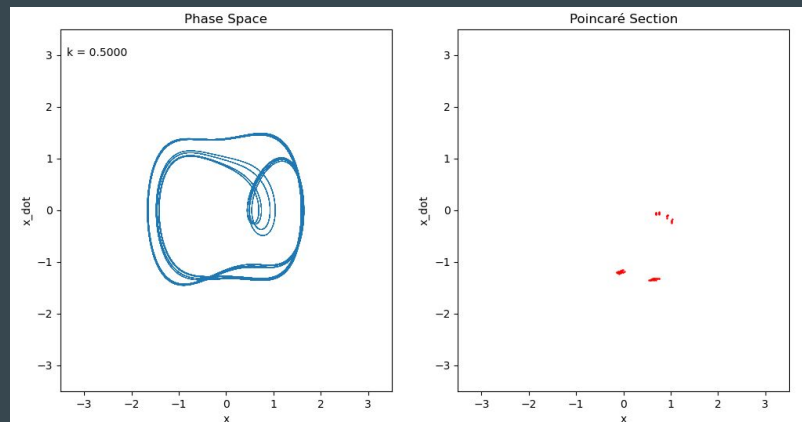
$$x'' + \gamma x' + \alpha x + \beta x^3 = k \cos \omega t$$

where x is position, γ is the damping coefficient, α is the restoring force constant, β is a non-linear correction to account for the potential, k is the drive magnitude, and ω is the drive frequency.



Intro II

- The regions of deterministic unpredictability in the Duffing Oscillator are highly dependant on small changes on the parameters of the differential equation.
- We can get a good idea about the behavior of the oscillator behavior from the projection onto phase space, particularly sampling once per driving period with Poincare diagrams. The number of points in the Poincare corresponds to the level of disorder.



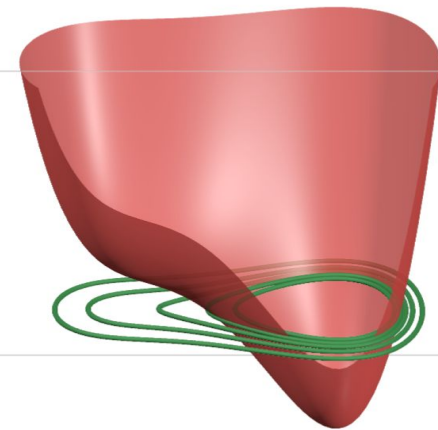
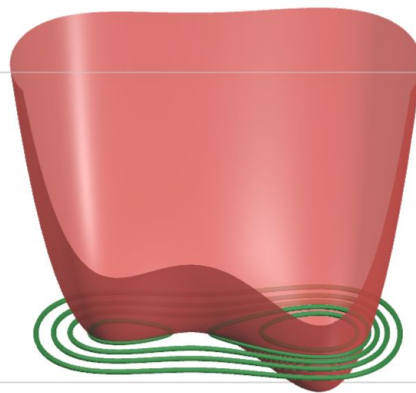
Methods I

- We modified the Duffing equation with a constant term, adding a linear slope to the potential function.

$$x'' + \gamma x' + \alpha x + \beta x^3 + C = k \cos \omega t$$

- When the slope is very large positive or negative, we should recover order in a predictable way, as we approach a single-well oscillator. As C approaches 0, we recover the original Duffing Oscillator.

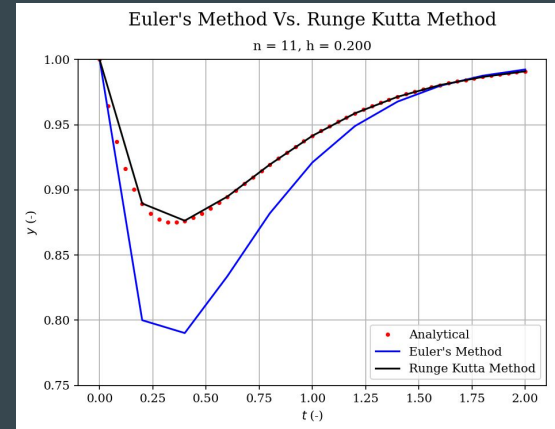
$C = .2$



$C = .9$

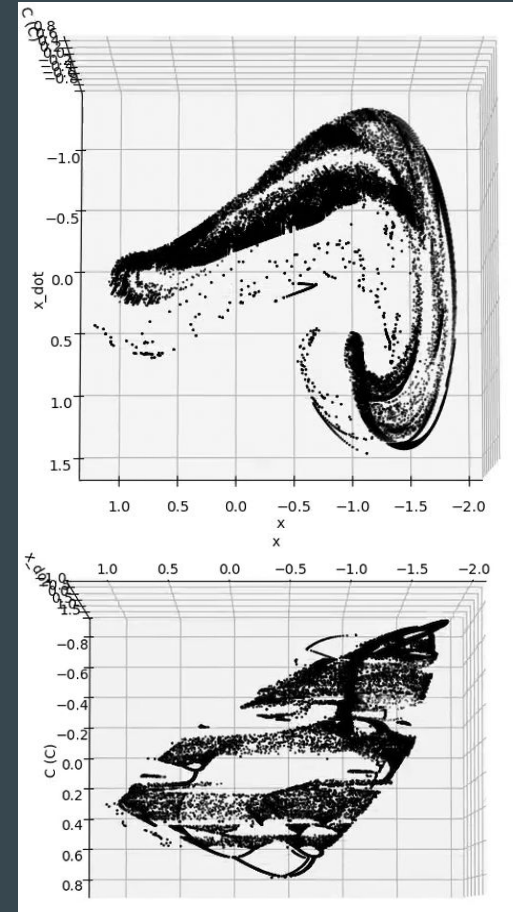
Methods II

- To simulate the Duffing Oscillator, we used numerical approximation through the Tsitouras 5/4 method. For all experiments unless otherwise noted, the time step was 0.005, and simulated from 0 to >5000. Poincare values were then indexed after calculations. As such there may be up to a 0.0025 error bar on all Poincare plot points



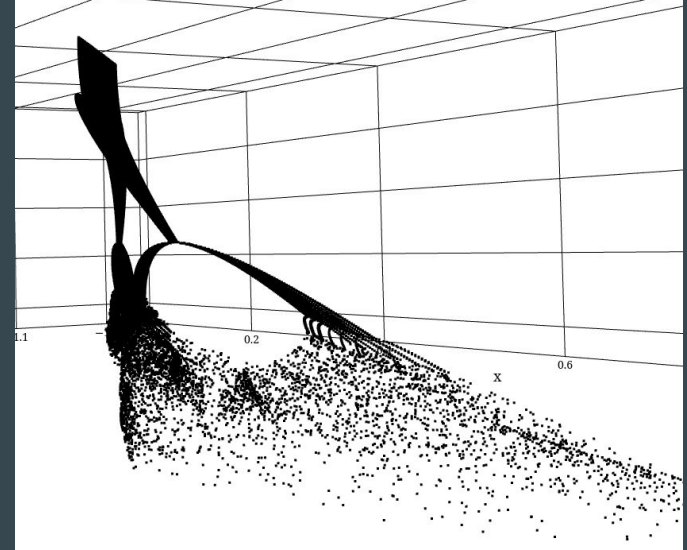
Methods III

- We first evaluated the Poincare plots for a variety of parameters. To see the regions of stability, we took the Poincare plot for each value of a given parameter, and plotted the scatter plot in three dimensions. This allows for a neatly contained plot that shows a bifurcation diagram on one plane, and the strange attractor on another.

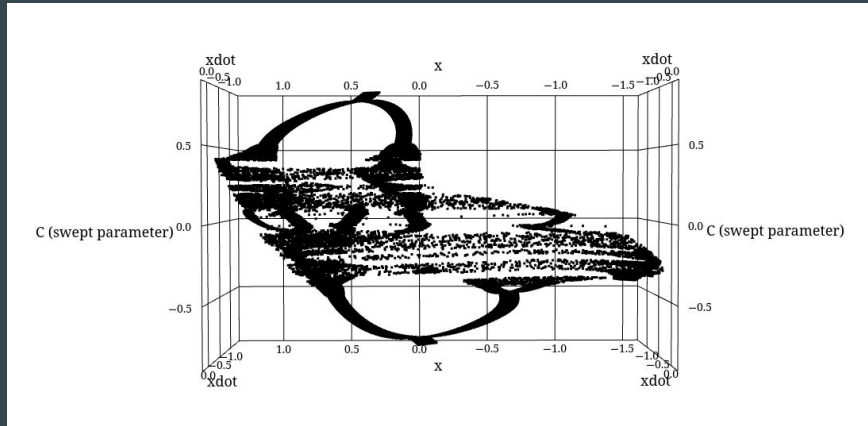


Methods IV

- With the Poincare plots from before, we can quite easily separate a given branch with simple statistical filters. This allows for a variety of analysis on the ‘nuclei’ of chaos, such as fractal dimension fitting, and rerunning with increased precision.



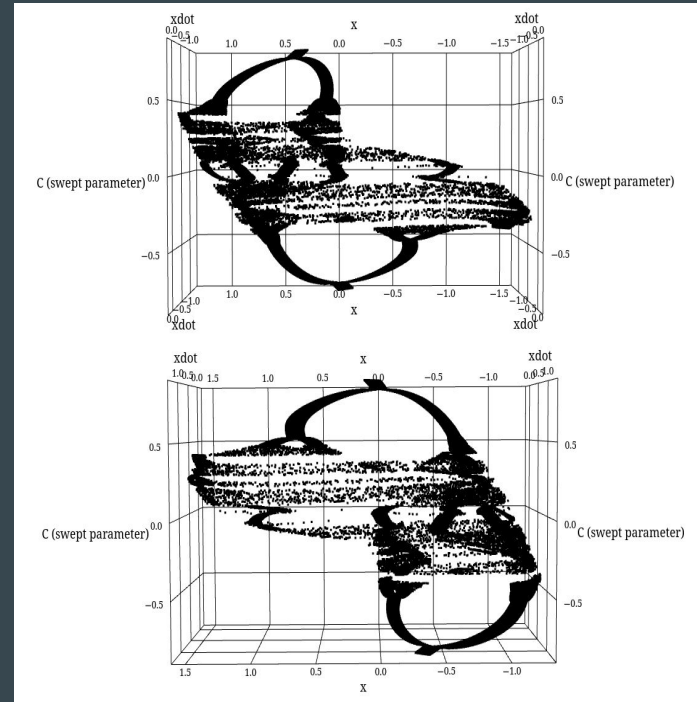
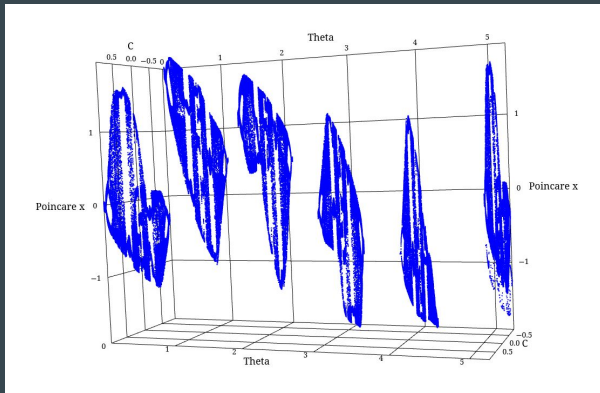
Results I - Asymmetry



- The asymmetry of the bifurcation plot here is odd. We might expect that the plot would be symmetric about $C=0$, since the potential is symmetric for $\pm C$. Instead we get something that has analogous structures on both sides of $C=0$ but is fully asymmetric.

Results II - Asymmetry

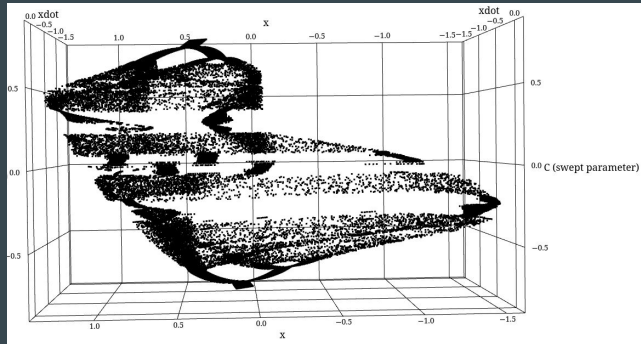
- Shifting the theta value within the Poincare sampling results in a mirroring about the x and xdot planes, which recovers symmetric behavior.
- If we slice even more values of theta, we assemble a classical sinusoidal plot



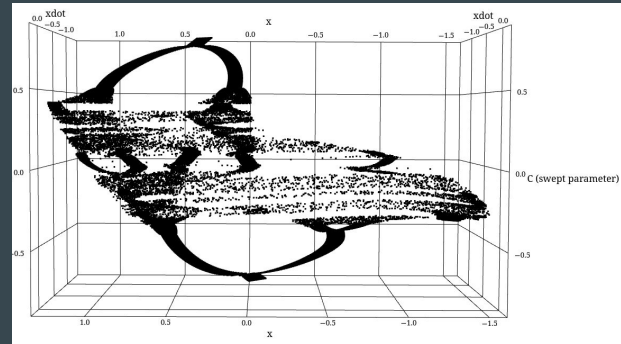
Results III - Multi-Variable Dependences

- Even within a single variable sweep, there is still strong dependance on initial conditions to determine chaotic regions.
- With two identical sweeps of C , a tiny change in k completely changes the regions of chaos

$k = 0.5$

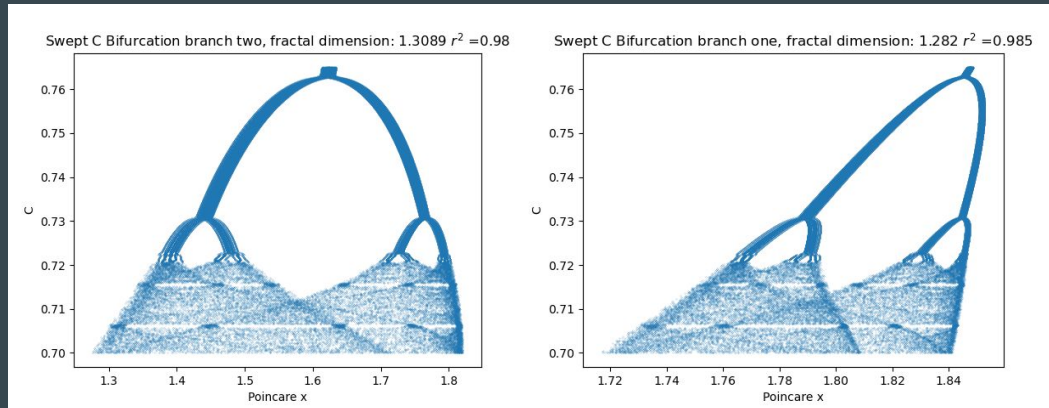


$k = 0.53$



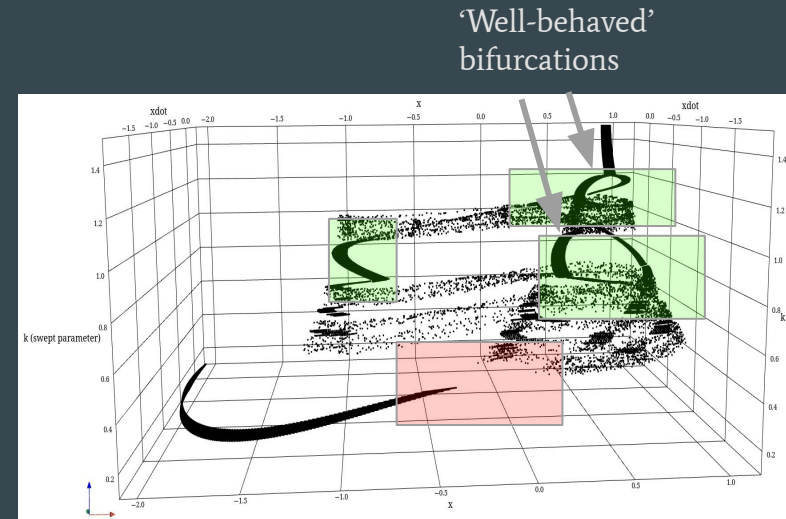
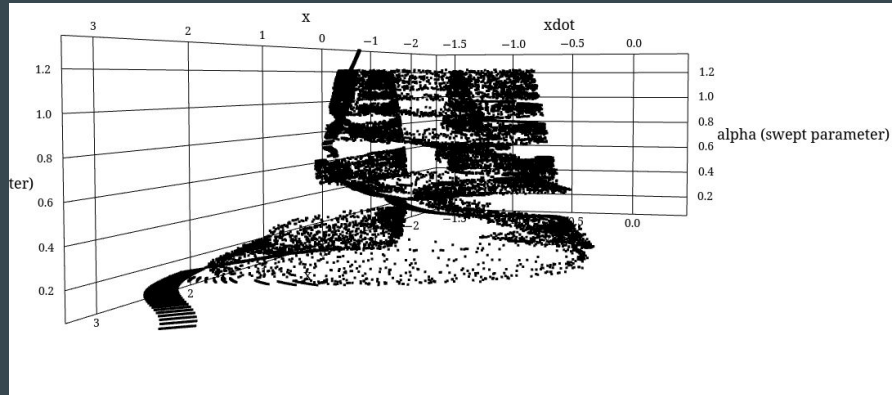
Results IV - Bifurcation

- We may look at the transition to chaos where C transitions from the single-well region of stability to the double-well region of instability.
- After separating the two main branches, we carried out standard box fitting, where we see a classic bifurcation plot which fits a fractal dimension of ~ 1.3 with r^2 of ~ 0.985 . This indicates that this is a classic fractal transition.



Results V - Other Chaotic Transitions

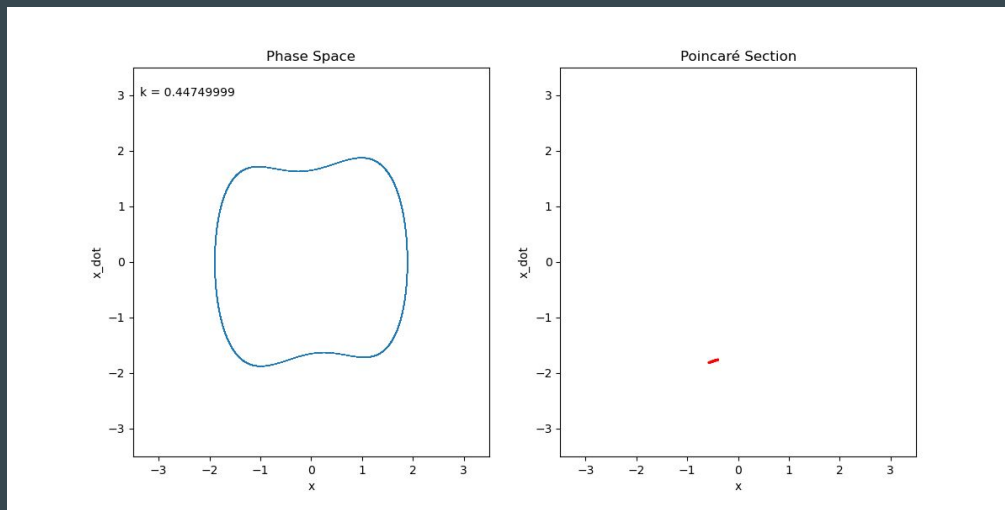
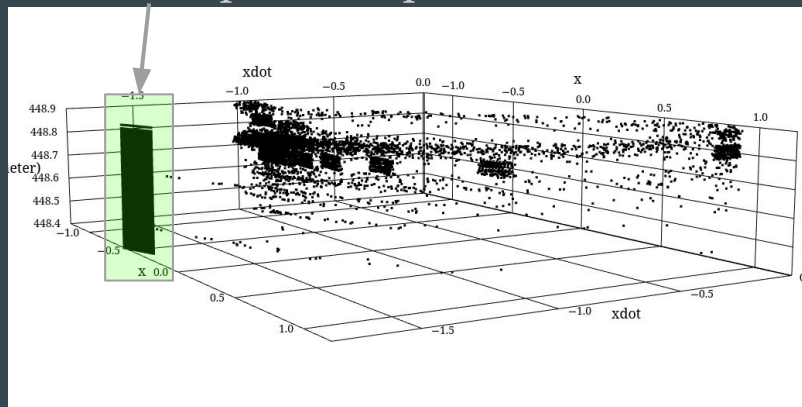
- A sweep of the k parameter shows a number of different types of transitions to and from chaos.
- Some, like the transition from $k=1$ to $k=1.4$ follow another well-behaved bifurcation series, others, like from $k=.1$ to $k=.5$ do not



Results VI - Uncalculatable Divergences

- Some jumps, like k from 0.4 to 0.5, are not represented as a bifurcation plot
- Vastly increasing numerical precision and zooming into that jump shows a brute-force attempt will not work. We numerically get nearly infinite jumps

‘Tail’ from previous plot



Conclusion

- Overall, the chaotic regions of the Duffing oscillator depend heavily on the initial parameters, and we can classically quantify only certain regions of stability.
- We got more insight into the shape of the bifurcations by the structure of the 3D plots, allowing us to analyze individual nodes on the plot, and fractally classify several regions of instability.
- By varying the parameter C , we see how chaos arises from the initial setup.

Thank You!

All code used is on the github <https://github.com/Thomaslund1/Phys231Final>



All 3D figures should be available from there, or on thomaslund.net/Chaos

